render_mermaid_blocks.py — companion guide

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Overview

`scripts/testing/render_mermaid_blocks.py` pre-processes a Markdown file so that every `` ```mermaid `` fenced block is replaced by a reference to a rendered **PNG** image, **before** the file is handed to pandoc. This closes the Constitution **§11.4.168** gap: pandoc has no mermaid filter and weasyprint cannot execute JavaScript, so without pre-rendering the diagram *source* would ship as literal text in the HTML/PDF exports.

It is invoked automatically by `scripts/testing/sync_all_markdown_exports.sh`; it can also be run standalone.

Prerequisites

- `python3` (stdlib only)
- `mmdc` (mermaid-cli) on `PATH` — `npm i -g @mermaid-js/mermaid-cli`
- Config: `scripts/testing/mermaid.config.json`

Usage

```
render_mermaid_blocks.py <input.md> <output.md> <png_dir>
[<cache_dir>]
```

- `<input.md>` — source (never modified)

- `<output.md>` — pre-processed copy; each fence → `![diagram](<abs-png>)`
- `<png_dir>` — where per-block PNGs are written
- `<cache_dir>` — optional content-addressed cache (sha256 of source+config); a cache hit reuses the PNG instead of re-rendering. The driver wrapper uses `<repo>/.mermaid-cache` (gitignored; regen mechanism = the wrapper, §11.4.77).

Prints `mermaid: rendered=N reused=M failed=K`.

Why PNG (not SVG)

Empirically (`§11.4.6`): weasyprint does **not** render mermaid's default-config SVG `<foreignObject>` text — it vanishes in the PDF. An `htmlLabels:false` SVG keeps native text but breaks `<br/>` labels. Rendering to PNG lets Chromium (inside `mmdc`) do the full rendering — every label feature — and weasyprint simply embeds the raster, which it can never fail to display. OCR of the rendered PDF page confirms readability (the §11.4.168 validation oracle). Proven across all 8 diagram types the spec uses.

Edge cases

- **A diagram fails to render** (syntax error): the script exits **1** and the output carries a `**[MERMAID RENDER FAILED — block #N]**` marker — NEVER raw source. The wrapper marks the file FAILED. Fix the diagram source (see `validate_mermaid_blocks.md`) — a render failure is a real defect, not a tooling glitch.
- **No mermaid fences**: the input is copied verbatim.

Related scripts

- `scripts/testing/validate_mermaid_blocks.py` — render-validate every diagram
- `scripts/testing/sync_all_markdown_exports.sh` — the export driver
- `scripts/testing/mermaid.config.json` — mermaid render config

Last verified: 2026-06-26 (rendered all 356 spec diagrams; 0 raw-source leaks into PDF confirmed via pdftotext + OCR).